

# Herald



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## CLAUDE.md

| Field          | Value                                                                                                                                                                                                                                                                                                                                                  |
|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Revision       | 6                                                                                                                                                                                                                                                                                                                                                      |
| Created        | 2026-05-15                                                                                                                                                                                                                                                                                                                                             |
| Last modified  | 2026-05-20                                                                                                                                                                                                                                                                                                                                             |
| Status         | active                                                                                                                                                                                                                                                                                                                                                 |
| Status summary | r5: added §107 End-user-usability covenant section restating the verbatim operator mandate at Herald level + ToC entry; ties to HERALD_CONSTITUTION.md §107 + inheritance-gate invariant I8a (paired with I8b/c on AGENTS.md + HERALD_CONSTITUTION.md).                                                                                                |
| Issues         | none                                                                                                                                                                                                                                                                                                                                                   |
| Issues summary | —                                                                                                                                                                                                                                                                                                                                                      |
| Fixed          | spec-path references (r2), pre-implementation-language update (r3), submodules + HRD-docs + codegraph-index enumeration (r4), §107 mandate restatement + I8a anchor (r5)                                                                                                                                                                               |
| Fixed summary  | aligned with HRD-009/HRD-009b/HRD-013/HRD-014 landing in the same commit; r4 closes the discoverability gaps observed during a fresh <code>/init</code> review; r5 closes the Herald-level explicit-restatement gap identified by the 2026-05-20 audit (Helix root + 10 submodules had the verbatim mandate; only Herald's three root docs lacked it). |
| Continuation   | bump again when first-implementation cycle completes HRD-010..HRD-012/HRD-016 live integrations.                                                                                                                                                                                                                                                       |

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This file provides guidance to Claude Code (claude.ai/code) when working with code in this repository.

## INHERITED FROM Helix Constitution (parent-discovery)

Herald is consumed as a submodule of a parent project that already carries the Helix Constitution submodule at <parent>/constitution/. Herald therefore does **NOT** keep its own copy. Locate the constitution from any nested depth by walking up parents until you find <ancestor>/constitution/Constitution.md, or by running the canonical helper:

```
CONST_DIR="$(bash "$(find . -type d -name constitution -print
    -quit 2>/dev/null)/find_constitution.sh)"
# or, more robustly, from any starting directory:
CONST_DIR="$(bash <ancestor>/constitution/find_constitution.sh)"
```

For standalone development of Herald (no parent project), clone the constitution alongside Herald:

```
git clone git@github.com:HelixDevelopment/HelixConstitution.git \
    $(dirname "$PWD")/constitution
```

Once located, all rules in <discovered>/CLAUDE.md and the <discovered>/Constitution.md it references apply unconditionally. Herald-specific rules below extend them — they MUST NOT weaken any inherited rule. When this file disagrees with the discovered constitution, the constitution wins.

Canonical: <https://github.com/HelixDevelopment/HelixConstitution>

### Read order on a cold start:

1. <discovered-constitution>/CLAUDE.md + Constitution.md — universal Helix rules. Inherited unconditionally.
2. <discovered-constitution>/AGENTS.md — agent guardrails (anti-bluff, no-guessing, paired mutations).
3. README.md — Herald overview + how-to.
4. This file (Herald-specific notes below).
5. docs/guides/HERALD\_CONSTITUTION.md — Herald's project-specific constitutional extensions.
6. docs/guides/CONSTITUTION\_INHERITANCE.md — operator/agent guide for the discovery contract + the inheritance gate.
7. docs/specs/mvp/specification.V3.md — mission spec (mostly TBD).

# Project status

Herald is in **first-implementation cycle (r1)** as of 2026-05-20. The Go scaffold has landed; live-integration work (HRD-010..HRD-012, HRD-016) is still open. The repo contains:

- `README.md` — mission, deployment model, inheritance contract, quickstart.
- `docs/specs/mvp/specification.V3.md` — MVP spec stub (substantive sections TBD).
- `docs/guides/HERALD_CONSTITUTION.md` — Herald's project constitution (extends Helix).
- `docs/guides/CONSTITUTION_INHERITANCE.md` — operator/agent guide for parent-discovery + gate semantics.
- `upstreams/` — Herald's mirror declarations (see below).
- `tests/test_constitution_inheritance.sh + _meta.sh` — paired inheritance gate (§1.1).
- `.gitignore` tuned for Go + `.DS_Store`.

Herald does **not** ship a `constitution/` submodule of its own; the parent project provides it. See `docs/guides/CONSTITUTION_INHERITANCE.md`.

**As of 2026-05-20** the Go scaffold has landed (first-implementation cycle r1). The repo now contains:

- `go.work` (gitignored locally; check in if the project wants reproducible workspaces — current convention: gitignored per spec §9.1).
- `commons/ (L0)` — `commons/types.go` with the full §11.0 Channel contract + `Subscriber/CloudEventEnvelope/TraceContext/Branding/ChannelID`; `commons/clock.go` Clock abstraction; `commons/uuidv7.go`; `commons/branding.go` per-flavor factory.
- `commons_prefix/` — §8.2 3-letter prefix algorithm.
- `commons_messaging/channels/null/` — full §11.14 `null://` sandbox adapter (working, tested).
- `commons_messaging/channels/tgram/` — Telegram adapter SCAFFOLD (HRD-011 open).
- `commons_messaging/dispatch/claude_code/` — Claude Code session-resolution + envelope formatter; `live claude --resume` invocation pending (HRD-012 open).
- `commons_storage/` — 9 SQL migrations (000001..000005) embedded via `// go:embed`; `pgx + River + Redis` wiring pending (HRD-010 open).
- `pherald/cmd/pherald/` — Cobra CLI; `pherald version` works end-to-end; other subcommands stubbed with HRD-NNN error pointers.
- `quickstart/` — Herald-specific Docker/Podman Compose + Dockerfile + `otel-config + .env.example` per §26.5 (HRD-008 open for operator validation). Migrated from `containers/quickstart/` 2026-05-20 when the `containers/` submodule was added.
- `containers/` — git submodule pointing at `git@github.com:vasic-digital/containers.git` (the `digital.vasic.containers` Go module — runtime auto-detection + on-demand container boot + lifecycle/health). Imported by Foundation tests + the `pherald doctor` subcommand to bring Postgres + Redis + OTel up on-demand.

**Build + test:** from the repo root:

```

go build ./commons/... ./commons_prefix/... ./
commons_messaging/... ./commons_storage/... ./
commons_constitution/... ./commons_infra/... ./
pherald/...
go test -race -count=1 ./commons/... ./commons_prefix/... ./
commons_messaging/... ./commons_storage/... ./
commons_constitution/... ./commons_infra/... ./
pherald/...

```

Tests pass on Go 1.25+ (verified on 1.26). Workspace is configured via `go.work` listing 7 Herald modules (`commons`, `commons_constitution`, `commons_infra`, `commons_messaging`, `commons_prefix`, `commons_storage`, `pherald`). The 9 Helix-stack submodules under `submodules/` are referenced via `replace` directives in the consuming modules' `go.mod`, not via `go.work`.

### Anti-bluff verification (run before any release tag or risky commit):

```

scripts/audit_antibluff.sh      # 3 invariants: §11.4 anchor +
                                tests + paired meta
scripts/codegraph_validate.sh   # CodeGraph index integrity (7
                                probes)
scripts/e2e_bluff_hunt.sh       # 14 end-to-end checks against
                                real services

```

CodeGraph index lives in `.codegraph/` (`codegraph.db` + `config.json`, both gitignored). Rebuild with `scripts/codegraph_setup.sh` when source layout changes; `codegraph_validate.sh` will FAIL otherwise.

`scripts/e2e_bluff_hunt.sh` is the canonical end-to-end smoke per Universal §11.4. It builds pherald, runs the full test suite, starts a real Gin server + hits every `/v1` route + asserts response bodies, boots a real Postgres container via the `containers/` submodule, runs the M2 integration tests against it, and graceful-shutdowns. ALL 14 invariants must PASS — a single FAIL means a real feature is broken for end users. At 2026-05-20 it reports 14/14 PASS.

When the user asks to "add a feature" the spec is the source of truth — find the relevant §, then the relevant module + package, then the relevant HRD-NNN if one is already open. New work opens a new HRD-NNN in `docs/Issues.md` per V3 §8.3 lifecycle.

Do not invent build/test commands beyond what `go test ./<module>/...` provides. Live-integration tests (Telegram bot, Claude Code session, real Postgres) require operator-supplied credentials — see `docs/CONTINUATION.md` for the live-test handoff prompt.

## End-user-usability covenant (Herald §107 / Helix §11.4 — MANDATORY ANTI-BLUFF)

### Forensic anchor — verbatim operator mandate:

"all existing tests and Challenges do work in anti-bluff manner - they MUST confirm that all tested codebase really works as expected! We had been in position that all tests do execute with success and all Challenges as well, but in reality the most of the features does not work and can't be used! This MUST NOT be the case and execution of tests and Challenges MUST guarantee the quality, the completion and

full usability by end users of the product! This **MUST BE** part of Constitution of our project, its CLAUDE.MD and AGENTS.MD if it is not there already, and to be applied to all Submodules's Constitution, CLAUDE.MD and AGENTS.MD as well (if not there already)!"

The bar for shipping any Herald feature is **NOT** "tests pass" — it is **"the end user of the flavor binary can actually use the feature."** Every PASS (unit, integration, gate, Challenge, smoke, e2e) **MUST** carry positive runtime evidence that the user-visible behaviour works. Metadata-only / configuration-only / "absence-of-error" / grep-only PASS are §11.4 PASS-bluffs and constitute critical defects regardless of how green the summary line looks. Canonical Herald authority: docs/guides/HERALD\_CONSTITUTION.md §107. Canonical Helix authority: <discovered>/Constitution.md §11.4 + §11.4.1..§11.4.16 and <discovered>/CLAUDE.md "MANDATORY ANTI-BLUFF COVENANT — END-USER QUALITY GUARANTEE". Canonical Herald evidence: scripts/e2e\_bluff\_hunt.sh (14 invariants against real services; ALL must PASS). Inheritance gate invariant **I8a** asserts this covenant anchor is present in this file.

## Mission (from the spec)

Ingesting system events and reliably fanning them out to multiple notification channels so every alert reaches the right destination without confusion.

The spec mentions "Integration into the Constitution" as a planned section — Herald is intended to plug into a larger governance/policy system that lives outside this repo. Treat that as a real constraint when designing interfaces, even though the details are TBD.

## Intended stack

Go, inferred from .gitignore (\*.test, go.work, go.work.sum, coverage.\*, profile.cov). When you scaffold, default to standard cmd/ + internal/ layout unless the user asks otherwise, and use go test ./.../go build ./.../go vet ./... — there is no project-specific tooling to override these yet.

## Multi-host mirror convention

upstreams/ contains one script per mirror host:

- GitHub.sh — primary: git@github.com:vasic-digital/Herald.git
- GitLab.sh — git@gitlab.com:vasic-digital/herald.git
- GitFlic.sh — git@gitflic.ru:vasic-digital/herald.git
- GitVerse.sh — git@gitverse.ru:vasic-digital/Herald.git

Each script exports a single UPSTREAMABLE\_REPOSITORY variable and is meant to be **sourced** (. upstreams/GitHub.sh), not executed for its output. The naming is intentional — capitalization matches the host's brand (GitFlic, GitVerse). Don't normalize these to lowercase or collapse them into one file; the per-host split is the interface other tooling (likely external mirror-push scripts) keys on.

When adding a new mirror, copy an existing script and change only the URL — keep the #!/bin/bash shebang, blank line, and export form identical so any consumer that greps these files keeps working.

## Inheritance gate (run before any commit touching root docs)

```
bash tests/test_constitution_inheritance.sh
      # 6 invariants (I1-I6), 9 checks
bash tests/test_constitution_inheritance_meta.sh # paired §1.1
      mutation proof
```

The gate inline-walks parents for <ancestor>/constitution/Constitution.md. I5 is split into I5a–d (one check per root doc that must declare parent-discovery: CLAUDE.md, AGENTS.md, docs/guides/HERALD\_CONSTITUTION.md, README.md). I6 forbids re-introducing <repo-root>/constitution/ or .gitmodules — the §104 invariant.

If either script fails, fix at root cause per Universal §11.4.4. Never silently accept the FAIL.

## Spec-change rule (load-bearing — docs/specs/mvp/specification.V3.md §"Specification documents")

Any modification to a file under docs/specs/ (any depth) triggers **mandatory comprehensive planning and implementation** of the implied changes — you may not edit the spec in isolation. This rule does **not** apply to creating or renaming files; for those, ask the operator what to do with the new path. Treat every spec edit as a project-wide ripple, not a doc tweak.

## Project conventions from the spec (apply when scaffolding)

These are declared in docs/specs/mvp/specification.V3.md and are easy to miss because no code enforces them yet:

- **Workable-item prefix:** HRD- (e.g. HRD-001). Use it for issues, status entries, fix logs.
- **Flavor binaries:** each Herald flavor ships as its own CLI binary, named <prefix>herald — pherald (Project Herald), sherald (System Herald), etc. Designed for CI / pipeline / cron / AI-agent invocation.
- **Layered shared code:** commons → commons\_messaging (level 1) → ... → flavor. Put new shared code in the **lowest layer that still makes sense**; flavors inherit upward. commons\_messaging owns the Telegram / Max / Slack / Email / Markdown-export integrations.
- **Messenger integration priority:** Telegram → Max → Slack (then Email, then Markdown/PDF/HTML export). Microsoft Teams, Lark, Discord, WhatsApp, Viber are explicitly later iterations — don't pre-implement.
- **Conversation diary:** every message in/out is appended to docs/herald/diary/main.md and re-exported to main.pdf + main.html in sync. Don't break this invariant when designing channel I/O.
- **Container stack:** Postgres (main DB) + Redis (in-memory) bundled via the containers submodule (<https://github.com/vasic-digital/containers>). All container names start with herald; all host ports start with 70XXX (70001, 70002, ...) to avoid collisions.
- **Credentials:** .env (git-ignored) with a committed .env.example. Resolution order: exported shell vars from .bashrc/.zshrc load first; .env is fallback only (never overrides shell exports). **For step-by-step setup of every supported messenger + LLM dispatcher**, see docs/guides/OPERATOR\_CREDENTIALS.md — comprehensive guide covering Postgres, Redis, Telegram (HRD-011), Claude Code (HRD-012), plus

reserved env-var names for planned channels (Slack, Email, Max, Teams, ...) and dispatchers. The guide includes an audit checklist to run before every commit.

- **Vendored SDKs:** any official/unofficial messenger SDK or API client we depend on goes in as a **git submodule**, e.g. `commons_messaging/sdk/telegram` or `commons_messaging/api/telegram` — not go get'd into `go.mod`.

## constitutable/ directory (parent-project extension hook)

The empty `constitutable/` directory at the repo root is intentional. Per the spec, a parent project may drop additional `Constitution.md` / `CLAUDE.md` / `AGENTS.md` (in `constitutable/`, `constitutable/<flavor>/`, `constitutable/<flavor>/<variant>/`, etc.) to layer extensions or overrides on top of the discovered `constitution/` submodule. Apply-order: `constitution/` submodule → `constitutable/` extensions → Herald's own docs. Do not delete the directory because it's empty.

## Documentation artefacts (PDF/HTML siblings)

`docs/guides/HERALD_CONSTITUTION.md` and `docs/guides/CONSTITUTION_INHERITANCE.md` each ship with a committed `.pdf` sibling. When you edit one of these Markdown files, the PDF goes stale — flag it; do not regenerate silently unless the operator asks.

The HRD-lifecycle docs in `docs/` also ship as PDF/HTML/DOCX quadruples: `Issues.md` (open HRDs per V3 §8.3), `Fixed.md` (closed-HRD log per §11.4.19 atomic migration), `Status.md` (status summary), `CONTINUATION.md` (live-test handoff prompt for operator-supplied credentials). The `*_Summary.md` variants are derived; do not hand-edit.

**Logo branding (added 2026-05-21).** Every tracked Markdown doc now leads with a centered Herald logo header (pandoc-friendly `<div align="center">` wrapping a `![Herald] (...) {width=96px height=96px}` image reference to `assets/logo/herald_logo_square_128.png`). The export pipeline propagates that logo into the HTML, PDF, and DOCX siblings:

- Logo source: `assets/logo/herald_logo.png` (1664x928 RGB master). Square + transparent variants live under `assets/logo/herald_logo_square_{32,64,128,256,512,1024}.png` (chroma-keyed white → alpha). `assets/logo/herald_logo.svg` is a vector wrapper around the 512px PNG. `assets/logo/print.css` carries print/screen styling for the HTML/PDF route.
- Injection: idempotent re-runnable via `python3 scripts/branding_inject_logo.py <md ...>` — skips submodules/, containers/, `constitutable/`, `docs/diary/`, `LICENSE`; respects YAML front-matter; computes the relative path per doc depth.
- Export: `bash scripts/export_docs.sh [<md>...]` regenerates HTML (pandoc), PDF (`--pdf-engine=weasyprint`), DOCX (pandoc-native) for every `.md` that already has at least one sibling artefact committed. Pass no args to regenerate everything.
- When you add a new `.md`, run the injector once; when you edit one whose sibling exports are tracked, run the exporter for that one file.

## Notes for future scaffolding

- `submodules/` holds 9 vendored Helix-stack modules (each its own `git@github.com:vasic-digital/<name>.git` repo): `auth`, `background`, `cache`, `config`, `database`, `eventbus`, `middleware`, `observability`, `recovery`. They are referenced via `replace` directives in the consuming Herald modules' `go.mod`, NOT via `go.work` (which only lists the 7 Herald-owned modules). Do not add them to `go.work`.
- The repo is in `main` branch and committed under "Milos Vasic" — no other contributors yet.
- `.claude/` exists but is empty; project-local Claude config can go there.
- `LICENSE` is present (do not overwrite without asking).
- `.DS_Store` is now git-ignored; do not re-add the previously-stray files.